

FELIX JIMENEZ

Work Authorization: US Citizen

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EXPERIENCE

Research Assistant

Texas A&M University

-  Dec. 2020-Present
-  College Station, TX
- Scalable Bayesian optimization, and Gaussian processes approximation.
 - TA for statistical learning for one semester.

Mathematical Statistician (Prev. Research Assistant)

National Institute of Standards and Technology

-  05/2018-09/2020
-  Boulder, CO
- Stats research: GAN performance on low dimensional multimodal data, hierarchical Gaussian processes for outlier detection.
 - Applied research: Bootstrap for quantum tomography, hierarchical Bayesian models for chemical measurements, permutation tests for glass forensics.
 - Became permanent employee after starting as contractor.

Research Assistant

University of Colorado Boulder

-  01/2015-05/2018
-  Boulder, CO
- Stats consultant for mechanical engineering and social science projects.
 - Programmed numerical simulations for systems of interacting quantum dipoles.

PUBLICATIONS

[1] F. Jimenez and M. Katzfuss, "Scalable Bayesian Optimization Using Vecchia Approximations of Gaussian Processes," Under Review.

[2] F. Jimenez, A. Koepke, M. Gregg, and M. Frey, "Generative Adversarial Network Performance in Low-Dimensional Settings," *Journal of Research of National Institute of Standards and Technology*, vol. 126, 2021.

[3] K. Tucker, B. Zhu, R. J. Lewis-Swan, J. Marino, F. Jimenez, J. G. Restrepo, and A. M. Rey, "Shattered time: Can a dissipative time crystal survive many-body correlations?" *New J.Phys.*, vol. 20, Dec. 2018.

ABOUT ME

I'm a second year PhD student in statistics working on scalable Bayesian optimization for high dimensional problems.

EDUCATION

Ph.D., Statistics

Texas A&M University

 2020-2024

 College Station, TX

M.S. / B.S., Applied Mathematics

University of Colorado Boulder

 2014 – 2018

 Boulder, CO

Courses

- Bayesian inference • Bayesian computation • probabilistic graphical models • NLP • measure theory • math stat • linear models • network science • numerical analysis • time series • spatial statistics

Community Engagement

- Mentoring statistics undergrad student.
- Stat department student government.

TECHNICAL SKILLS

Programming

- Languages: Python, R
- Libraries: PyTorch, BoTorch, NumPy, Pyro
- Visualization and writing: ggplot, LaTeX.

Stats & ML

- Bayesian optimization, Bayesian statistics
- Gaussian processes, generative adversarial networks